SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Computer Science and Engineering

ASSESSMENT TOOL 2 – EXPERIMENTATION

Course Code:	ITA03	**Course Title:**	Mobile computing
CO Assessed:	CO2	**Assessment:**	Assessment Tool 2 – EXPERIMENTATION
Weightage:	30 % of CIA	**Total Marks:**	100 marks
CO2 Statement:	*Compare mobile communication technologies, including multiple access techniques and network evolution, based on performance and applications.*		
Student Name:	P.Ranjithkumar	**Reg. No.:**	192411119
Section / Batch:	2024	**Date:**	13.10.2026

Code of Conduct

I P.Ranjithkumar (192411119)_____ (Name / Reg. No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: P.Ranjithkumar_____

Instructions

- This test contains FIVE questions.
- Duration: 60 minutes (homework)

Faculty In-charge	Course Coordinator	Head of Department
Signature: _____	Signature: _____	Signature: _____
Date: _____	Date: _____	Date: _____

1. Figma-Based Mobile IP Communication Workflow Analysis

Aim

To design a Figma prototype showing how Mobile IP works when a Mobile Node moves from its Home Network to a Foreign Network.

Components

- Mobile Node (MN): A mobile device that changes networks.
- Home Agent (HA): Keeps track of the mobile node's current location.
- Foreign Agent (FA): Helps visiting devices in traditional Mobile IPv4.
- Correspondent Node (CN): The device communicating with the mobile node.
- Care-of Address (CoA): Represents the mobile node's current location.
- Registration: Updates the Home Agent with the current location.
- Packet Tunneling: Forwards packets from the Home Agent to the Care-of Address.

Workflow

Correspondent Node (CN) → Home Agent (HA) → IP Tunnel → Foreign Agent/Care-of Address → Mobile Node (MN).

Analysis

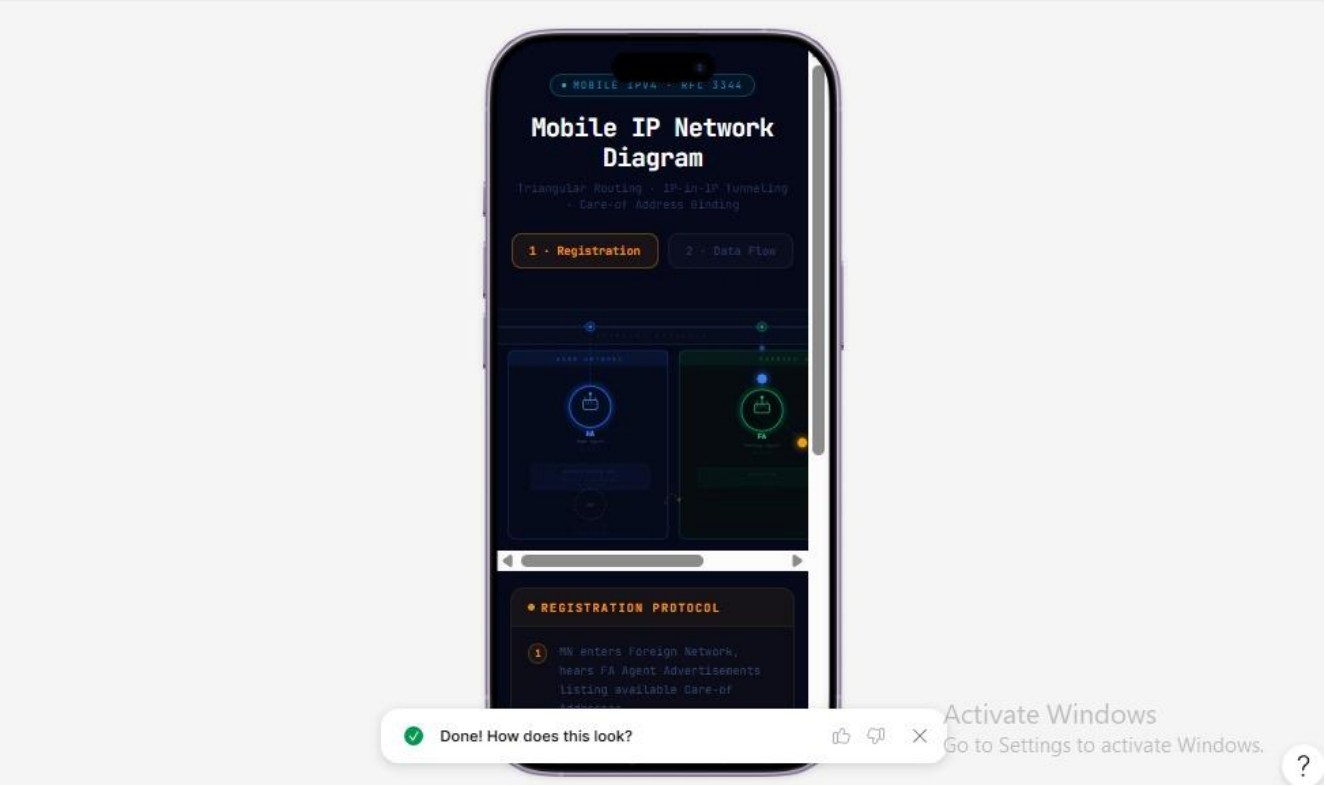
- Delay can occur during registration, address configuration, and handoff.
- Packet loss may occur while the mobile node changes networks.
- Frequent movement can increase delay and reduce communication performance.
- Fast handoff, packet buffering, and route optimization can help reduce latency.

Procedure

1. Open Figma and create a new design file. 2. Draw boxes for CN, HA, FA, CoA, and MN. 3. Connect the components with arrows. 4. Show registration and packet tunneling. 5. Add another screen showing handoff. 6. Export the design as PNG or PDF.

Result

The Mobile IP communication workflow is designed and analyzed.



2. Comparative Network Architecture Design Using Figma

Aim

To compare Cellular Wireless Networks and Mobile Ad Hoc Networks (MANET) in a smart city environment.

Theory

A cellular network uses base stations and fixed infrastructure. A MANET allows mobile devices to communicate directly or forward data for other devices without fixed infrastructure.

Comparison

Feature	Cellular Network	MANET
Infrastructure	Required	Not required
Routing	Managed network routing	Distributed routing
Scalability	Supports many users with suitable capacity	Depends on network conditions
Fault tolerance	Base station failure may affect coverage	Alternate routes may be available
Mobility	Handoff between cells	Routes change when nodes move
Applications	Mobile services and smart cities	Disaster and emergency networks

Advantages and Limitations

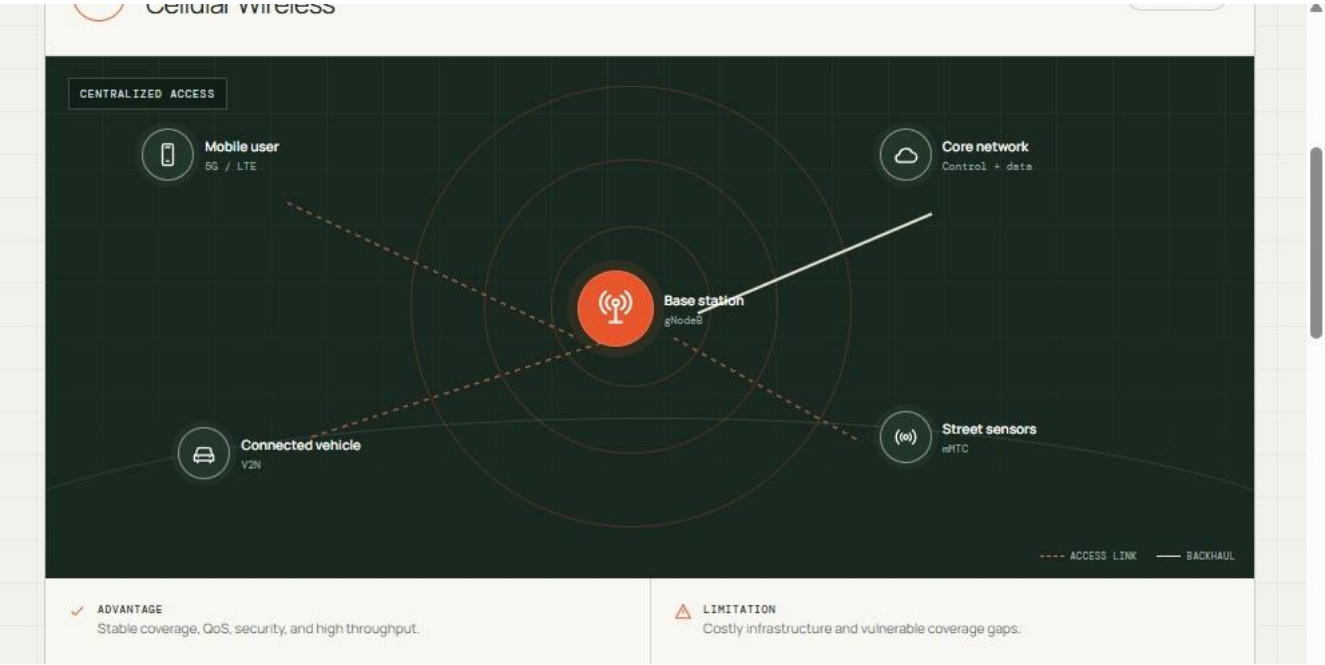
- Cellular advantages: wide coverage and managed connectivity.
- Cellular limitation: depends on infrastructure and coverage.
- MANET advantages: flexible and quick to deploy.
- MANET limitations: routes may break when nodes move.

Procedure

1. Open Figma. 2. Draw a cellular network with a base station and mobile devices. 3. Draw MANET nodes connected by wireless links. 4. Add arrows, labels, advantages, and limitations. 5. Export the design as PNG or PDF.

Result

The two network architectures are compared and analyzed.



3. Routing Protocol Visualization and Analysis

Aim

To visualize and compare a proactive routing protocol and a reactive routing protocol.

Protocols

- OLSR (Optimized Link State Routing): A proactive protocol that maintains routes in advance.
- AODV (Ad hoc On-Demand Distance Vector): A reactive protocol that discovers routes when needed.

Workflow

OLSR: Neighbor Discovery → Periodic Topology Updates → Route Available → Data Transmission.

AODV: Route Request (RREQ) → Route Reply (RREP) → Route Established → Data Transmission.

Comparison

Feature	OLSR	AODV
Route discovery	Maintains routes in advance	Discovers routes when needed
Route maintenance	Periodic updates	Route errors and rediscovery
Routing overhead	Periodic control traffic	Traffic during discovery and maintenance
Initial delay	Usually low if a route exists	May be higher during discovery
High mobility	Frequent topology updates	Frequent route rediscovery

Analysis

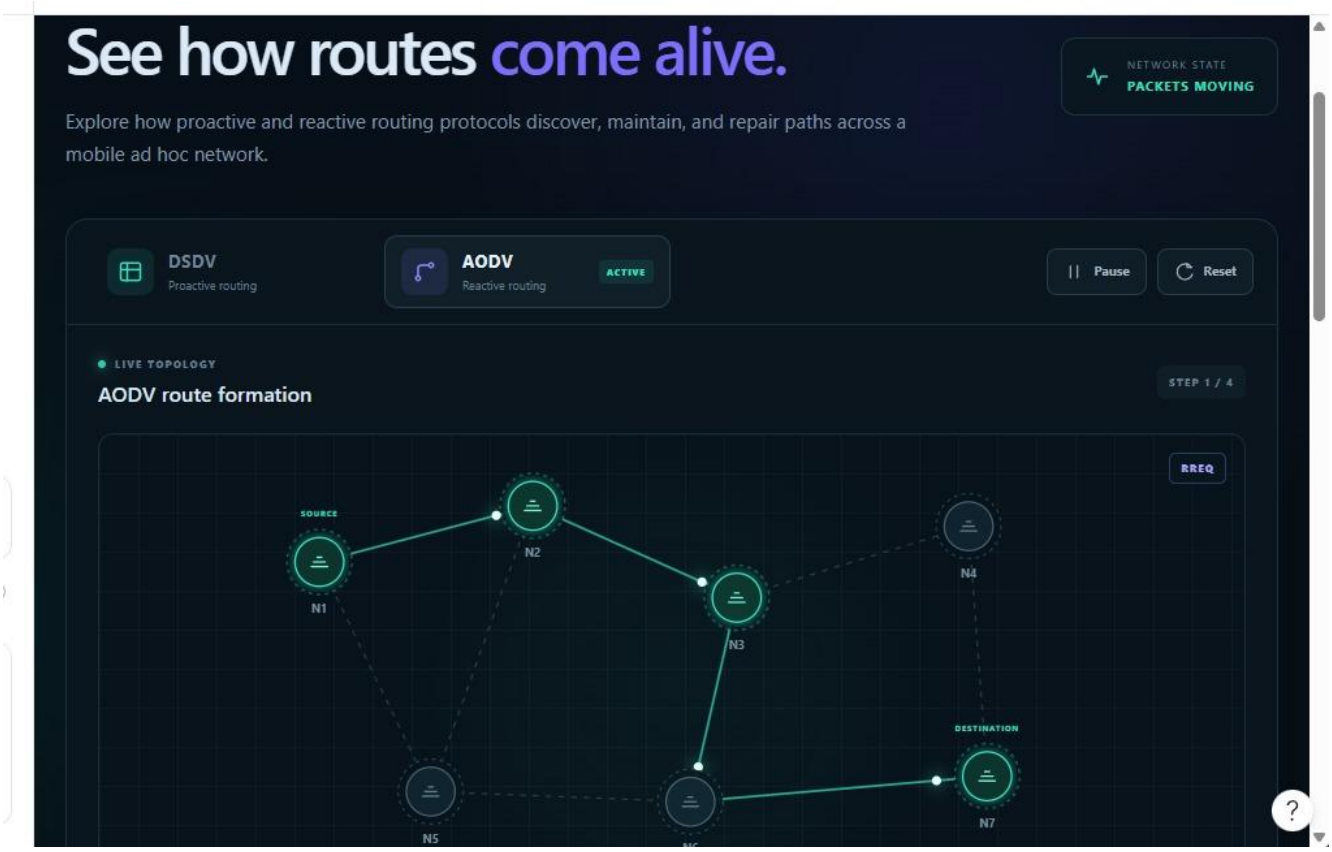
- OLSR can work well when communication is frequent because routes are already available.
- AODV can reduce periodic control traffic when communication is occasional.
- High mobility can affect both protocols.
- The better protocol depends on network size, traffic, and node movement.
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Procedure

1. Open Mermaid Live Editor at https://mermaid.live/. 2. Create a flowchart for OLSR. 3. Create a flowchart for AODV. 4. Add a comparison table. 5. Export the diagrams as images.

Result

OLSR and AODV are visualized and compared based on route discovery, overhead, delay, and mobility.



4. Mobile IP Application Design for Smart Healthcare

Aim

To design a Smart Healthcare Mobile IP system connecting doctors, ambulances, hospitals, and patients while they move between wireless networks.

Components

- Hospital server: Stores and provides healthcare information.
- Home Agent: Tracks the mobile device's registered location.
- Foreign Network: Provides connectivity in a new network area.
- Doctor, ambulance, and patient devices: Use the system to communicate.

Workflow

Ambulance/Doctor/Patient Device → Wireless Network → Home Agent → Hospital Server.

Analysis

- Mobile IP helps maintain communication when a device changes networks, but does not guarantee zero interruption.
- Handoff can cause delay and packet loss.
- Security risks include unauthorized registration and interception of medical data.
- Encryption, authentication, packet buffering, and fast handoff can improve reliability.

Screens to Design

- Login screen for doctors, ambulances, hospitals, and patients.
- Healthcare dashboard showing patient status and network status.
- Mobility screen showing the current network, CoA, and registration status.
- Emergency screen showing ambulance location and communication status.

Procedure

1. Open Figma. 2. Design the four screens. 3. Add suitable icons and labels. 4. Show a network handoff scenario. 5. Connect screens using prototype navigation. 6. Export the design as PDF or PNG.

Result

A Smart Healthcare Mobile IP prototype is designed to demonstrate mobility challenges and possible reliability improvements.



5. Wireless Network Decision Dashboard Using GitHub

Aim

To create a dashboard that recommends Mobile IP, Cellular Networks, or MANET based on network requirements.

Parameters

- Node mobility
- Infrastructure availability
- Network size
- Routing complexity
- Energy consumption
- Reliability requirements

Working

The user selects the network requirements. The system analyzes the selected parameters and recommends a suitable network with reasons.

Example Recommendations

Scenario	Suggested Network
Large city with many mobile users	Cellular Network
A device moving between IP networks while retaining its home address	Mobile IP
Emergency communication without fixed infrastructure	MANET

Procedure

1. Create a folder named Wireless-Network-Dashboard. 2. Create index.html, style.css, and script.js. 3. Add input fields for the six parameters. 4. Write JavaScript rules to recommend a network. 5. Display the recommendation and reasons. 6. Upload the files to GitHub. 7. Add screenshots and a README file.

Result

A wireless network decision dashboard is developed to recommend a suitable network architecture.

